

Binomial expansion - Set 2

TMUA practice questions by topic. Specification MM2.4. 4 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 The expansion of $\left(2x - \frac{1}{x^2}\right)^9$ contains a term in x^k for each of which of the following values of k ?
- I $k = 0$
 - II $k = 3$
 - III $k = -6$
- A none of them
- B I only
- C II only
- D III only
- E I and II only
- F I and III only
- G II and III only
- H I, II and III
- 2 The coefficient of x^2 in $(1+x)^n + (1+x)^{n+1}$ is 36. Find n .
- A 4
- B 8
- C 6
- D 5
- E 9
- F 7

3 n is a positive integer. Which of the following statements is/are true?

I The coefficient of x^n in $(1+x)^{2n}$ is $\binom{2n}{n}$.

II The sum of the coefficients in the expansion of $(1-x)^n$ is 0.

III $(1+x)^n \geq 1+nx$ for all $x \geq 0$.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

4 How many terms in the expansion of $(x+y+z)^4$ (after collecting like terms) are there?

A 16

B 15

C 10

D 5

E 12

F 81